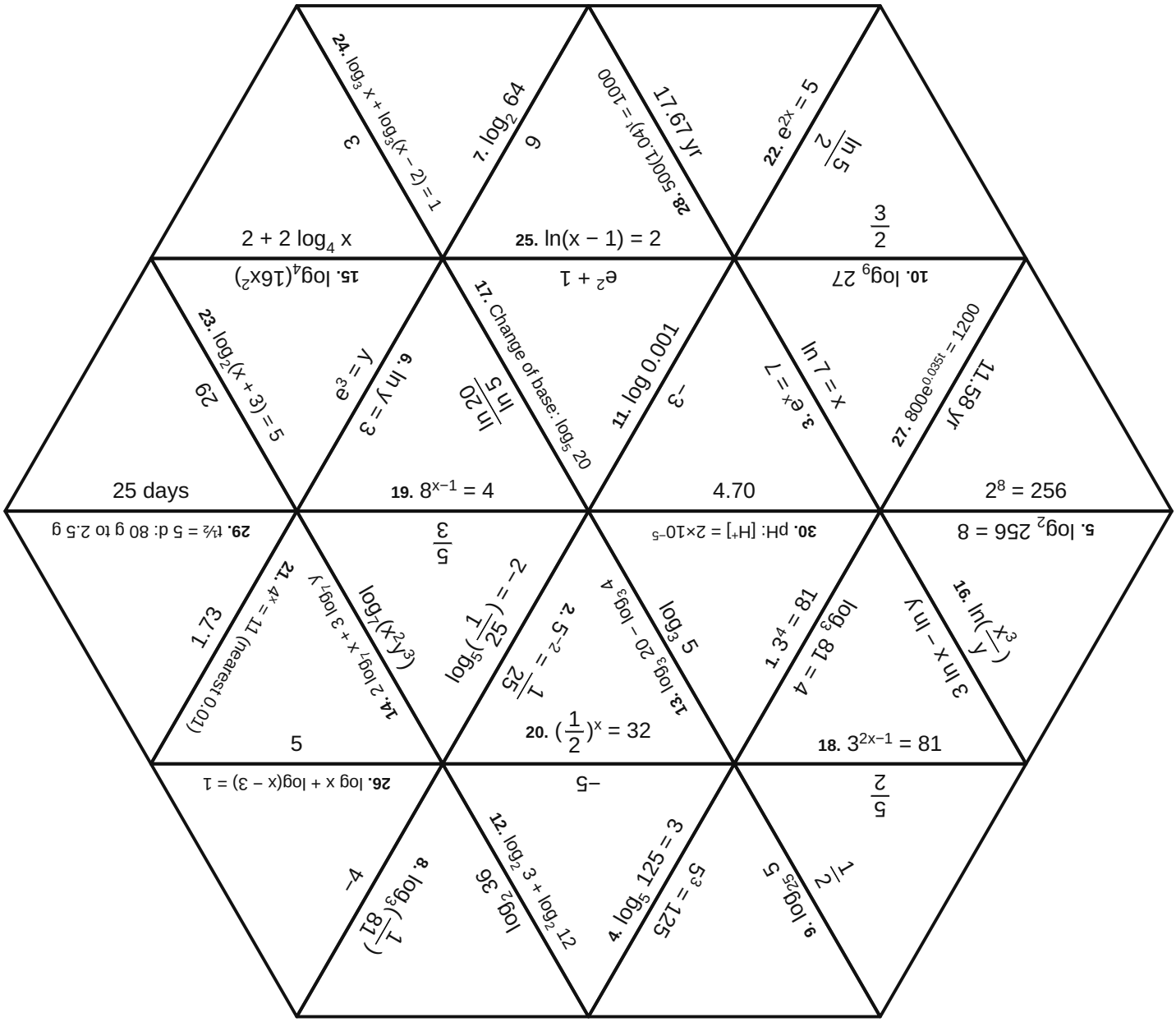
Exp & Log Functions Tarsia: Answer Key

Solved hexagon. Each shared edge pairs a problem with its answer.



Answer List

Exp & Log Functions Tarsia · all 30 matches

#	Problem	Answer (on the matching edge)
1	$3^4 = 81$	$\log_3 81 = 4$
2	$5^{-2} = \frac{1}{25}$	$\log_5(\frac{1}{25}) = -2$
3	$e^x = 7$	$\ln 7 = x$
4	$\log_5 125 = 3$	$5^3 = 125$
5	$\log_2 256 = 8$	$2^8 = 256$
6	$\ln y = 3$	$e^3 = y$
7	$\log_2 64$	6
8	$\log_3(\frac{1}{81})$	-4
9	$\log_{25} 5$	$\frac{1}{2}$
10	$\log_9 27$	$\frac{3}{2}$
11	$\log 0.001$	-3
12	$\log_2 3 + \log_2 12$	$\log_2 36$
13	$\log_3 20 - \log_3 4$	$\log_3 5$
14	$2\log_7 x + 3\log_7 y$	$\log_7(x^2y^3)$
15	$\log_4(16x^2)$	$2 + 2\log_4 x$
16	$\ln(\frac{x^3}{y})$	$3\ln x - \ln y$
17	Change of base: $\log_5 20$	$\frac{\ln 20}{\ln 5}$
18	$3^{2x-1} = 81$	$\frac{5}{2}$
19	$8^{x-1} = 4$	$\frac{5}{3}$
20	$(\frac{1}{2})^x = 32$	-5
21	$4^x = 11$ (nearest 0.01)	1.73
22	$e^{2x} = 5$	$\frac{\ln 5}{2}$
23	$\log_2(x + 3) = 5$	29
24	$\log_3 x + \log_3(x - 2) = 1$ Note: $x = -1$ is extraneous (logs of negatives)	3
25	$\ln(x - 1) = 2$	$e^2 + 1$
26	$\log x + \log(x - 3) = 1$ Note: $x = -2$ is extraneous (logs of negatives)	5
27	$800e^{0.035t} = 1200$	11.58 yr
28	$500(1.04)^t = 1000$	17.67 yr
29	$t_{½}$ = 5 d: 80 g to 2.5 g	25 days
30	pH: $[H^+] = 2\times10^{-5}$	4.70